A Hermite-moment proof of the recurrence for binary matrices with prescribed margins

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Abstract

Let $a(n)$ count labeled $3n$-by-6 binary matrices whose row sums are 2 and whose column sums are n, with $a(0) = 1$. We prove the specific third-order polynomial recurrence conjectured by Kauers and Koutschan for $b(n) = a(n)/\binom{3n}{n}$, for every integer $n \geq 0$. The count is a normalized sixth Gaussian moment of a Hermite polynomial. Gaussian integration by parts reduces the shift in the polynomial degree to three moments, and an elementary elimination gives the prescribed coefficients. The proof includes the endpoint $n = 0$ and requires no assumption based on finite numerical agreement. A small, exact, standard-library Python verifier checks its algebraic identities; separate combinatorial computations check the initial terms and a longer prefix.

1 The counting problem and the theorem

All row and column positions are labeled. Thus two matrices obtained from one another by a permutation are counted separately whenever their entries differ. For $n \in \mathbb{Z}_{\geq 0}$, let $a(n)$ count the matrices described in the abstract. The empty matrix contributes $a(0) = 1$. Set

$$b_n = b(n) = \frac{a(n)}{\binom{3n}{n}}.$$

Each row selects one of the fifteen pairs of distinct columns, so

$$a(n) = [x_1^n \cdots x_6^n] \left(\sum_{1 \leq i < j \leq 6} x_i x_j \right)^{3n}. \tag{1}$$

The ordinary power records the ordered sequence of rows; no quotient by a permutation group is taken.

Theorem 1. *For every integer $n \geq 0$,*

$$\begin{aligned} &(n+3)^4(62n^2 + 217n + 191)b_{n+3} - 6P(n)b_{n+2} \\ &\quad - 4(2n+3)Q(n)b_{n+1} \\ &\quad + 6000(n+1)^2(2n+1)(2n+3)(62n^2 + 341n + 470)b_n = 0, \end{aligned} \tag{2}$$

where

$$\begin{aligned} P(n) &= 5084n^6 + 68634n^5 + 383756n^4 + 1137319n^3 \\ &\qquad + 1884032n^2 + 1653960n + 601185, \\ Q(n) &= 31372n^5 + 313720n^4 + 1227805n^3 \\ &\qquad + 2354425n^2 + 2220988n + 827860. \end{aligned}$$

Its initial values are $b_0 = 1$, $b_1 = 30$, and $b_2 = 4530$. In particular, $a(1) = 90$, $a(2) = 67950$, and $a(3) = 90291600$.

The leading coefficient in (2) is strictly positive on $n \geq 0$, so these three initial values determine the entire sequence. This is a proof of the specified recurrence, not merely of the existence of some polynomial-coefficient recurrence.

2 Related work and the scope of the contribution

The matrix sequence is OEIS A172572 [2]. The recurrence in Theorem 1 is exactly Conjecture 6 of Kauers and Koutschan [1, p. 19]. Their Theorem 8 proves D-finiteness; that result alone does not prove the conjectured operator. Their paper also supplies multivariate summation and diagonal formulations. We checked the coefficient transcription against the journal article and its latest listed arXiv revision, version 2 of 24 April 2023.

The connection between Hermite products and pairings is classical; see Azor, Gillis and Victor [3]. Paris [4, Section 3] explicitly studies sixth Hermite moments and gives finite hypergeometric expressions. General polynomial-product techniques also appear in Hounkonnou, Belmehdi and Ronveaux [5], and general Hermite-product integrals in Dehesa, Moreno-Balcázar and Toranzo [6]. The moment representation itself is therefore not claimed as new.

The contribution here is the explicit three-moment reduction and its elimination to the particular recurrence. A bounded literature search on 28 October 2026 found no earlier proof of this exact recurrence. The OEIS entry still calls it conjectural. This observation does not establish global priority. The mathematical proof below is self-contained and does not depend on a novelty claim.

3 A Gaussian moment representation

Let Z be a standard real Gaussian random variable, with density $(2\pi)^{-1/2}\exp(-z^2/2)$. Define the probabilists' Hermite polynomials by

$$\exp(tz - t^2/2) = \sum_{m\geq 0} \mathrm{He}_m(z)\frac{t^m}{m!}. \tag{3}$$

Differentiating this identity gives

$$\mathrm{He}_m' = m\,\mathrm{He}_{m-1}, \qquad \mathrm{He}_{m+1} = z\,\mathrm{He}_m - m\,\mathrm{He}_{m-1}, \qquad \mathrm{He}_m(-z) = (-1)^m\,\mathrm{He}_m(z). \tag{4}$$

The first two polynomials are $\mathrm{He}_0 = 1$ and $\mathrm{He}_1 = z$.

Lemma 2. *For every integer $n \geq 0$, with $h_n = (2n)!/(n!)^5$,*

$$\frac{\mathbb{E}[\mathrm{He}_n(Z)^6]}{(n!)^6} = \frac{a(n)}{(3n)!}, \qquad b_n = h_n\mathbb{E}[\mathrm{He}_n(Z)^6]. \tag{5}$$

Proof. Multiplying six copies of (3) and taking expectations gives

$$\begin{aligned}\mathbb{E}\prod_{i=1}^{6}\exp(t_iZ - t_i^2/2) &= \exp\left(\frac{(\sum_i t_i)^2 - \sum_i t_i^2}{2}\right)\\ &= \exp\left(\sum_{i<j} t_it_j\right).\end{aligned}$$

Extract the coefficient of $t_1^n\cdots t_6^n$. On the right, only the $3n$-th power in the exponential contributes, since each monomial of $\sum_{i<j} t_it_j$ has degree 2. Equation (1) now proves the first identity. The second follows by dividing $a(n)$ by $\binom{3n}{n}$. Coefficient extraction is legitimate coefficientwise: every coefficient is a polynomial in Z and has finite Gaussian moments. □

4 Reduction to three mixed moments

Fix an integer $n \geq 1$ and write $y(z) = \mathrm{He}_n(z)$ and $u(z) = \mathrm{He}_{n-1}(z)$. Then

$$y' = nu, \qquad u' = zu - y. \tag{6}$$

The latter identity follows from (4) when $n \geq 2$; for $n = 1$ it reads $0 = z - z$, so it remains valid. Define

$$M_{j,k} = \mathbb{E}[Z^j y(Z)^{6-k} u(Z)^k], \qquad j \geq 0, \quad 0 \leq k \leq 6,$$

and put $I_k(n) = M_{0,k}$. Dependence on the fixed n is suppressed in $M_{j,k}$. Parity gives $M_{j,k} = 0$ whenever $j + k$ is odd.

Lemma 3. *The moments satisfy*

$$(k-1)M_{j+1,k} + jM_{j-1,k} + n(6-k)M_{j,k+1} - kM_{j,k-1} = 0. \tag{7}$$

Terms with zero coefficients are omitted; set $M_{-1,k} = 0$ when needed. In particular, $I_0(n) = 5nI_2(n)$. For $j \geq 1$ and $k \neq 1$,

$$M_{j,k} = \frac{kM_{j-1,k-1} - n(6-k)M_{j-1,k+1} - (j-1)M_{j-2,k}}{k-1}. \tag{8}$$

For $j \geq 1$ the remaining case has the formula

$$M_{j,1} = \frac{jM_{j-1,0} + 5njM_{j-1,2} + 20n^2 M_{j,3}}{j+1+4n}. \tag{9}$$

These equations express every required $M_{j,k}$ as a rational linear combination of $I_2(n)$, $I_4(n)$ and $I_6(n)$.

Proof. For any polynomial f, integration by parts gives $\mathbb{E}[f'(Z)] = \mathbb{E}[Zf(Z)]$; the boundary term is zero because a polynomial times $\exp(-z^2/2)$ tends to zero at both ends of the real axis. Apply this to $f = z^j y^{6-k} u^k$ and use (6) to obtain (7). Its case $j = 0, k = 1$ gives $I_0 = 5nI_2$, and replacing j by $j - 1$ gives (8).

To obtain (9), combine (7) at $(j, 0)$, $(j + 1, 1)$ and $(j, 2)$. In detail,

$$\begin{aligned} M_{j+1,0} &= jM_{j-1,0} + 6nM_{j,1}, \\ M_{j+1,2} &= 2M_{j,1} - 4nM_{j,3} - jM_{j-1,2}, \\ 0 &= (j+1)M_{j,1} + 5nM_{j+1,2} - M_{j+1,0}. \end{aligned}$$

Substitution gives the claimed formula. Its same-degree term $M_{j,3}$ is reduced by (8), which decreases j. Thus the reduction terminates at $j = 0$. At that level, odd moments vanish and $I_0 = 5nI_2$. All denominators are nonzero for $n \geq 1$ in their stated domains. □

For brevity put

$$\begin{aligned} F(n) &= 102n^4 + 204n^3 + 145n^2 + 43n + 5, \\ G(n) &= 62n^2 + 93n + 36, \\ H(n) &= 62n^2 + 31n + 5. \end{aligned} \tag{10}$$

Proposition 4. *For $n \geq 1$ and $v_n = (I_2(n), I_4(n), I_6(n))^{\mathsf{T}}$,*

$$v_{n+1} = A(n)v_n, \tag{11}$$

where

$$A(n) = \begin{pmatrix} \dfrac{3nF(n)}{(n+1)(2n+1)} & -\dfrac{9n^4 G(n)}{2(n+1)(2n+1)} & -\dfrac{5n^6}{2(n+1)} \\ \dfrac{nH(n)}{2n+1} & -\dfrac{60n^4}{2n+1} & 0 \\ 5n & 0 & 0 \end{pmatrix}. \tag{12}$$

Proof. Increasing n by one replaces (y,u) by $(Zy-nu,y)$. Expanding therefore gives

$$\begin{aligned} I_2(n+1) &= M_{4,0} - 4nM_{3,1} + 6n^2M_{2,2} - 4n^3M_{1,3} + n^4I_4(n),\\ I_4(n+1) &= M_{2,0} - 2nM_{1,1} + n^2I_2(n),\\ I_6(n+1) &= I_0(n) = 5nI_2(n). \end{aligned} \tag{13}$$

For an explicit check of the algebra, set $A = I_2(n)$, $B = I_4(n)$, $C = I_6(n)$. Lemma 3 yields

$$M_{1,1} = \frac{5n((3n+1)A - 3n^2B)}{2n+1}, \qquad M_{1,3} = \frac{3}{2}(A - nB),$$
$$M_{2,0} = 5nA + 6nM_{1,1}, \qquad M_{4,0} = 3M_{2,0} + 6nM_{3,1},$$
$$M_{2,2} = \frac{(18n^2+2n-1)A + (-18n^3+6n^2)B}{2n+1}.$$

The remaining expression is

$$M_{3,1} = \frac{5n^2}{4(n+1)(2n+1)}\Big((84n^2+96n+18)A + (-74n^3-39n^2+26n)B - (2n^4+n^3)C\Big).$$

Substitute these expressions into (13) and collect the coefficients of A, B, C. The result is (12). These displayed finite substitutions provide a directly checkable certificate for every matrix entry. □

5 Eliminating the auxiliary moment

Keep $h_n = (2n)!/(n!)^5$, and introduce

$$c_n = 5n^3h_nI_4(n), \qquad d_n = 5n^5h_nI_6(n) \qquad (n \geq 1).$$

Since $I_0 = 5nI_2$, the scaled state is

$$(b_n, c_n, d_n)^{\mathsf{T}} = h_n \operatorname{diag}(5n, 5n^3, 5n^5)v_n.$$

The factorial ratio

$$\frac{h_{n+1}}{h_n} = \frac{2(2n+1)}{(n+1)^4}$$

transforms (12) into

$$\begin{pmatrix} b_{n+1}\\ c_{n+1}\\ d_{n+1}\end{pmatrix} = \begin{pmatrix} \dfrac{6F(n)}{(n+1)^4} & -\dfrac{9nG(n)}{(n+1)^4} & -\dfrac{5n(2n+1)}{(n+1)^4}\\ \dfrac{2H(n)}{n+1} & -\dfrac{120n}{n+1} & 0\\ 10(2n+1)(n+1) & 0 & 0\end{pmatrix}\begin{pmatrix} b_n\\ c_n\\ d_n\end{pmatrix}. \tag{14}$$

There is a direct formula for the third coordinate, including its initial value. By Lemma 2,

$$d_n = 5n^5\frac{h_n}{h_{n-1}}b_{n-1} = 10n(2n-1)b_{n-1} \qquad (n \geq 1). \tag{15}$$

Consequently, the first two rows of (14) say

$$(n+1)^4b_{n+1} = 6F(n)b_n - 9nG(n)c_n - 50n^2(4n^2-1)b_{n-1}, \tag{16}$$
$$(n+1)c_{n+1} = 2H(n)b_n - 120nc_n. \tag{17}$$

Both hold for every integer $n \geq 1$.

Proof of Theorem 1. Write (16) at $n+1$ and substitute (17). This gives

$$(n+2)^4 b_{n+2} = 6F(n+1)b_{n+1} + 1080nG(n+1)c_n \\ - (18G(n+1)H(n) + 50(n+1)^2(4(n+1)^2-1))b_n.$$

Multiply by $G(n)$ and replace $9nG(n)c_n$ using (16). After moving all terms to the left, one obtains

$$\begin{aligned} 0 = {} & G(n)(n+2)^4 b_{n+2} \\ & - 6(G(n)F(n+1) - 20G(n+1)(n+1)^4)b_{n+1} \\ & + (18G(n)G(n+1)H(n) + 50G(n)(n+1)^2(4(n+1)^2-1) \\ & \qquad - 720G(n+1)F(n))b_n \\ & + 6000n^2(4n^2-1)G(n+1)b_{n-1}. \end{aligned} \tag{18}$$

This elimination uses multiplication and substitution only. The following two polynomial identities follow by expansion:

$$G(n)F(n+1) - 20G(n+1)(n+1)^4 = P(n-1), \tag{19}$$

$$18G(n)G(n+1)H(n) + 50G(n)(n+1)^2(4(n+1)^2-1) \\ -720G(n+1)F(n) = -4(2n+1)Q(n-1). \tag{20}$$

Set $n = m+1$ in (18), use

$$G(m+1) = 62m^2 + 217m + 191, \qquad G(m+2) = 62m^2 + 341m + 470,$$

and factor $4(m+1)^2 - 1 = (2m+1)(2m+3)$. Equations (19)–(20) then give exactly (2) at index m.

The derivation held for every $n \geq 1$, so it proves the assertion for every $m \geq 0$. In particular, no undefined He_{-1} is used: the lowest moment pair is $(\mathrm{He}_1, \mathrm{He}_0)$, and (15) at $n = 1$ uses $b_0 = 1$. The initial values are evaluated below, completing the theorem. □

6 Initial values and independent counting checks

At $n = 1$, $y = Z$, $u = 1$, and the Gaussian moments $\mathbb{E}[Z^4] = 3$, $\mathbb{E}[Z^2] = 1$ give

$$v_1 = (3,1,1)^{\mathsf{T}}, \qquad (b_1, c_1, d_1) = (30, 10, 10).$$

Equation (14) at $n = 1$ gives $(b_2, c_2, d_2) = (4530, 2340, 1800)$, and its first row at $n = 2$ gives $b_3 = 1074900$. Multiplication by $\binom{3n}{n}$ yields

$$a(1) = 90, \qquad a(2) = 67950, \qquad a(3) = 90291600.$$

For an additional endpoint check, the residual at index zero is

$$81 \cdot 191 \cdot 1074900 - 6 \cdot 601185 \cdot 4530 - 12 \cdot 827860 \cdot 30 + 6000 \cdot 3 \cdot 470 = 0.$$

The beginning of the sequence is as follows.

n	$a(n)$	b_n
0	1	1
1	90	30
2	67950	4530
3	90291600	1074900
4	154700988750	312527250
5	306407299538340	102033732780

Two independent exact coefficient methods provide additional checks. First, if $D(r_1, \dots, r_6)$ counts ordered rows with residual column sums r_i, then

$$D(r) = \sum_{i<j,\ r_i,r_j>0} D(r - e_i - e_j), \qquad D(0) = 1,$$

with $D(r) = 0$ if $\sum_i r_i$ is odd or $\max_i r_i > \frac{1}{2}\sum_i r_i$. The latter condition says a column cannot appear more often than the number of remaining rows. Permuting the residual margins preserves the count, so sorted tuples can be used as cache keys while retaining the multiplicity of each chosen column pair.

Second, set

$$T_n(t) = \sum_{r=0}^{\lfloor n/2 \rfloor} \frac{n!}{r!(n-2r)!} t^r, \qquad t_{n,k} = [t^k] T_n(t)^6.$$

The identity $2e_2 = e_1^2 - \sum_i x_i^2$ in six variables gives

$$a(n) = \frac{(3n)!}{2^{3n}(n!)^6} \sum_{k=0}^{6\lfloor n/2 \rfloor} (-1)^k \frac{(6n-2k)!}{(3n-k)!}\, t_{n,k}. \tag{21}$$

Indeed, choose k quadratic terms in the binomial expansion of $(e_1^2 - \sum_i x_i^2)^{3n}$, distribute them as $(r_1, \dots, r_6)$, and extract the remaining exponents $n - 2r_i$ from e_1^{6n-2k}. The resulting factorial factors give (21). All its factorial arguments are nonnegative in the stated range.

The supplied implementation uses exact integers. The residual-margin method and (21) agree for $0 \le n \le 15$. The latter computes through $n = 100$ independently of the recurrence; the recurrence residual is zero for $0 \le n \le 97$. These finite checks are corroboration, not a step in the all-index proof.

7 Exact verification and reproducibility

Every essential formula used in the proof is displayed above. The supplement provides a second way to check the finite algebra:

1. `src/verify_proof.py` implements rational functions in a formal variable over $\mathbb{Q}$ using polynomial coefficient arrays and the Python standard library. It derives the transition matrix from (8)–(9), verifies the scaling in (14), and checks both polynomial identities (19)–(20) by exact symbolic equality. It also checks conversion to the published unnormalized recurrence.

2. `src/independent_count.py` carries out the two exact coefficient computations described in the preceding section.

3. `src/audit_moments.py` independently constructs Hermite polynomials and integrates polynomial products with exact Gaussian moments, checking the displayed transition on small degrees.

The first verifier checks identities as rational functions, rather than testing them at finitely many values of n. The analytic bridges, the combinatorial interpretation, and the nonvanishing denominator conditions are established by the proofs above. This is not a claim of formal verification in a proof assistant.

From the root of the supplied project, the standard-library checks are reproduced with Python 3.9 or later:

```
python src/verify_proof.py
python src/independent_count.py --max-n 100 --dp-max-n 15
python src/audit_moments.py
```

An optional independent symbolic derivation, `src/derive_moments.py`, uses SymPy. The paper can be rebuilt with a conventional LaTeX installation:

```
pdflatex -interaction=nonstopmode -halt-on-error \
  -jobname=research_paper -output-directory=output paper/main.tex
```

Run that command twice to resolve the references. The included `README.md` describes the saved reports and release check.

Authorship and verification scope. This manuscript and its scripts were developed by GPT-6 Astra, an AI system, and checked through separate derivations and exact computations. No external peer review or proof-assistant certification is asserted. The result proved is Theorem 1 on its full stated domain.

References

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